

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q18-1086

Baldwyn 100 kV  
May 23, 2020

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1086. The Interconnection Customer proposes a 87.8 MW solar photovoltaic facility interconnecting at the Baldwyn 100 kV substation in the Ohio area. The request was received on October 25, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 354 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$102,925,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

The Grand Ave-S O'Neil 100 KV is overloaded to 151.7% of its 136 MVA emergency rating for the loss of the Camden Wire Substation-Adirondack 100 KV. The project contributes 5.0% DFAX (4.4 MW) to this overload.

Exxonmobil Baton Rouge Turbine Generator-Clear Lake Power 100 KV: 150.0% loading (125 MVA) following the outage of the Swan Valley Substation-Trego 500 KV; DFAX = 14.4%.

Clear Lake Power-Latin Quarter Substation 100 KV: 148.4% loading (119 MVA) following the outage of the Trego-Swan Vly 500 KV; DFAX = 14.4%.

For the loss of Taylorville-Baldwn 100 KV, the Bishop Creek 3-Clapham 100 KV loads to 148.3% of its 139 MVA rating. Project DFAX: 7.1%.

For the loss of Baldwyn-Taylorville 100 KV Line, the Eustis-Bishop Crk 3 100 KV loads to 147.5% of its 136 MVA rating. Project DFAX: 7.1%.

Grand Ave Substation-Ray Braswell Substation 100 KV line: 145.7% loading (131 MVA) following the outage of the Adirondack-Camden Wire 100 KV L; DFAX = 5.0%.

The N Trail-Eustis 100 KV L is overloaded to 144.4% of its 134 MVA emergency rating for the loss of the Baldwyn-Taylorville 100 KV Line. The project contributes 7.1% DFAX (6.3 MW) to this overload.

For the loss of Camden Wire-Adirondack 100 KV line, the Ray Braswell-Clapham 100 KV loads to 144.1% of its 123 MVA rating. Project DFAX: 5.0%.

For the loss of Eastown Road-Sterling 100 KV, the E Side Lima-S O'Neil 100 KV loads to 118.7% of its 115 MVA rating. Project DFAX: 7.5%.

The Latin Quarter Substation-Ocoee 100 KV is overloaded to 115.6% of its 139 MVA emergency rating for the loss of the Baldwyn-Hay Canyon 100 KV L. The project contributes 7.5% DFAX (6.6 MW) to this overload.

For the loss of Baldwyn-Hay Canyon 100 KV L, the Ocoee-Jeffrey 100 KV loads to 109.7% of its 135 MVA rating. Project DFAX: 7.5%.

For the loss of Hay Canyon-Baldwyn 100 KV Line, the Bonanza Substation-Jeffrey Substation 100 KV loads to 109.6% of its 138 MVA rating. Project DFAX: 7.5%.

For the loss of Trego-Swan Vly 500 KV, the Exxonmobil Baton Rouge Turbine Generator-Buffer 100 KV loads to 105.9% of its 155 MVA rating. Project DFAX: 14.4%.

Soda Lake-Bonanza 100 KV: 104.9% loading (135 MVA) following the outage of the Baldwin-Hay Canyon 100 KV; DFAX = 7.5%.

For the loss of Brookline-Stockton Cogen 100 KV, the Uilk Wind Llc-N Trail 100 KV loads to 103.9% of its 126 MVA rating. Project DFAX: 5.2%.

For the loss of Baldwin-Hay Canyon 100 KV L, the Soda Lake-Hopkinsville 100 KV loads to 102.2% of its 133 MVA rating. Project DFAX: 7.5%.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Grand Ave-S O'Neil 100 KV                                         | Rebuild line with higher-capacity conductor | \$10,858,000  | 100% (\$10,858,000)   |
| Exxonmobil Baton Rouge Turbine Generator-Clear Lake Power 100 KV  | Rebuild line with higher-capacity conductor | \$13,229,000  | 100% (\$13,229,000)   |
| Clear Lake Power-Latin Quarter Substation 100 KV                  | Rebuild line with higher-capacity conductor | \$7,349,000   | 100% (\$7,349,000)    |
| Bishop Crk 3-Clapham Substation 100 KV                            | Rebuild line with higher-capacity conductor | \$13,525,000  | 100% (\$13,525,000)   |
| Eustis-Bishop Crk 3 100 KV                                        | Rebuild line with higher-capacity conductor | \$7,330,000   | 100% (\$7,330,000)    |
| Grad Ave-Ray Braswell 100 KV L                                    | Rebuild line with higher-capacity conductor | \$13,557,000  | 100% (\$13,557,000)   |
| Eustis-North Trail 100 KV                                         | Rebuild line with higher-capacity conductor | \$8,829,000   | 100% (\$8,829,000)    |
| Ray Braswell-Clapham 100 KV                                       | Rebuild line with higher-capacity conductor | \$5,444,000   | 100% (\$5,444,000)    |
| East Side Lima-S O'Neil 100 KV                                    | Rebuild line with higher-capacity conductor | \$1,257,000   | 100% (\$1,257,000)    |
| Latin Quarter Substation-Ocoee 100 KV                             | Rebuild line with higher-capacity conductor | \$8,798,000   | 100% (\$8,798,000)    |
| Ocoee-Jeffrey 100 KV                                              | Reconductor line                            | \$1,994,000   | 100% (\$1,994,000)    |
| Bonanza-Jeffrey 100 KV L                                          | Reconductor line                            | \$898,000     | 100% (\$898,000)      |
| Exxonmobil Baton Rouge Turbine Generator-Buffer Substation 100 KV | Reconductor line                            | \$2,323,000   | 100% (\$2,323,000)    |
| Soda Lake-Bonanza 100 KV                                          | Reconductor line                            | \$2,346,000   | 100% (\$2,346,000)    |
| Uilk Wind Llc-North Trail 100 KV                                  | Reconductor line                            | \$1,082,000   | 100% (\$1,082,000)    |
| Hopkinsville-Soda Lake 100 KV Line                                | Reconductor line                            | \$4,106,000   | 100% (\$4,106,000)    |
|                                                                   | Total allocated to project                  |               | \$102,925,000         |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.